Lec 4

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Amplitude Scaling in Signal Analysis — Complete Notes

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1. What Amplitude Scaling Actually Is

Amplitude scaling (also called **amplitude modification**, **gain**, or **magnitude scaling**) is one of the most basic — and most important — operations you can perform on a signal. It means multiplying every value of a signal by a **constant factor**, without touching the time axis at all.

If $x(t)$ is a continuous-time signal, amplitude scaling produces a new signal:

$$y(t) = A \cdot x(t)$$

If $x[n]$ is a discrete-time signal, the same idea applies:

$$y[n] = A \cdot x[n]$$

Here, A is a real (or sometimes complex) constant called the **scaling factor** or **gain**. It is a single number that multiplies *every sample* of the signal identically — it does not depend on t or n. This is the key distinguishing feature: amplitude scaling is a **memoryless, time-invariant, linear** operation.

Why it matters: almost every real system you'll study — amplifiers, attenuators, volume controls, AGC (automatic gain control) circuits, ADC/DAC normalization stages — is fundamentally performing amplitude scaling, sometimes combined with other operations. Understanding it precisely is the foundation for understanding gain, decibels, signal-to-noise ratio, and linear system theory.

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2. The Three Basic Signal Operations (Where Amplitude Scaling Fits)

In signal analysis, there are three elementary operations performed on the *independent variable* and *dependent variable* of a signal. Amplitude scaling is the operation on the **dependent variable** (the amplitude axis), as opposed to the other two which act on the **independent variable** (the time axis):

Operation	Formula	What it changes
Time Shifting	$y(t) = x(t - t_0)$	Moves the signal left/right along the time axis
Time Scaling	$y(t) = x(at)$	Compresses/expands the signal along the time axis
Amplitude Scaling	$y(t) = A \cdot x(t)$	Stretches/compresses the signal along the *vertical* (amplitude) axis

This distinction matters because amplitude scaling and time scaling are frequently confused by beginners — time scaling changes *how fast* the signal plays out, while amplitude scaling changes *how loud/strong* it is, without altering timing at all.

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3. Types of Amplitude Scaling by the Value of A

The behavior of $y(t) = A \cdot x(t)$ depends entirely on the value of the constant A. This is the most important sub-classification in this topic.

3.1 Amplification ($|A| > 1$)

When $|A| > 1$, every sample of the signal is made **larger in magnitude**. The signal's peak-to-peak swing increases. This is what a volume-up control, an audio amplifier, or a gain stage in an op-amp circuit does. Example: $A = 2$, $x(t)$ ranges from -1 to $+1$ $y(t)$ ranges from -2 to $+2$.

3.2 Attenuation ($0 \;\; |A| \;\; 1$)

When $|A|$ is a positive fraction less than 1, every sample shrinks in magnitude. This is what a volume-down control or a resistive attenuator (voltage divider) does. Example: $A = 0.5$, $x(t)$ ranges from -1 to $+1$ $y(t)$ ranges from -0.5 to $+0.5$.

3.3 Inversion ($A = -1$)

When $A = -1$, the signal is flipped upside down (reflected about the time axis) but its magnitude is unchanged. This is called **amplitude inversion** or **phase inversion by** 10 in the context of sinusoidal signals — every peak becomes a trough and vice versa. Example: $x(t) = (t)$ $y(t) = -(t) = (t +)$.

3.4 Negative Scaling with Magnitude Change ($A \;\; -1$ or $-1 \;\; A \;\; 0$)

A combination of inversion **and** amplification/attenuation. For instance $A = -$ both flips the signal and triples its magnitude.

3.5 Identity ($A = 1$)

Trivial case — the signal is unchanged. Included here because it is the boundary case that separates "amplification" from "attenuation," and it's important in proofs (e.g., showing a system is linear, $A = 1$ must reproduce the original signal exactly).

3.6 Zero Scaling ($A = 0$)

The signal is completely erased — $y(t) = 0$ for all t. This is a degenerate but mathematically valid case, often used to model a "muted" channel or a switch in the "off" position.

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4. Graphical Effect of Amplitude Scaling

This is best understood visually: amplitude scaling **stretches or compresses the waveform vertically** while leaving every point on the time axis exactly where it was.

- The x**-axis (time) values are untouched** — if $x(t)$ had a feature at $t = 5$, $y(t)$ still has its corresponding feature at $t = 5$.
- Only the y**-axis (amplitude) values are multiplied**.
- Zero-crossings of the signal are **preserved** (except in the degenerate $A = 0$ case) — if $x(t_0) = 0$, then $y(t_0) = A \cdot 0 = 0$ too. This is a very useful fact: amplitude scaling never creates or removes a zero-crossing, never shifts *where* peaks occur in time, and never changes the signal's *shape* — it only changes the shape's vertical size.

4.1 Worked Example

Let $x(t)$ be a triangular pulse with peak value 4 at $t = 0$, sloping down to 0 at $t = 2$. For $A = 0.25$: $y(t)$ is the same triangular shape, still zero at $t = 2$, but now peaking at 1 instead of 4 at $t = 0$. The pulse "footprint" on the time axis is identical; only its height changed.

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5. Effect on Signal Power and Energy

This is the part of the topic most often tested, because it requires care with the **square** in the power/energy formulas.

5.1 Energy Signals

The energy of a continuous-time signal is defined as:

$$E_x = {}_{-} |x(t)|^2 t$$

For the scaled signal $y(t) = A \cdot x(t)$:

$$E_y = \int_{-} |A \cdot x(t)|^2 t = A^2 \int_{-} |x(t)|^2 t = A^2 \cdot E_x$$

Key result: energy scales with the SQUARE of the amplitude scaling factor, not the factor itself. If you double the amplitude ($A = 2$), the energy goes up by a factor of **4**, not 2.

5.2 Power Signals

Similarly, for power signals, average power is defined as:

$$_x = \mathrm{m}\, 12 \int_{-} |x(t)|^2 t$$

Applying the same scaling:

$$_y = A^2 \cdot {}_x$$

Same conclusion: power also scales by A^2.

5.3 Why This Matters in Practice

This A^2 relationship is the reason **decibels (dB) use a factor of 20 for amplitude/voltage ratios but a factor of 10 for power ratios** (see Section 6) — it's directly a consequence of power being proportional to the square of amplitude.

5.4 Worked Numeric Example

Suppose $x(t)$ has energy $E_x = 10$ Joules (in the signal-processing sense of the term). If we scale it by $A =$:

$$E_y = {}^2 \; 10 = \;\; 10 = 0$$

The energy increases **ninefold**, even though the amplitude only tripled.

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6. Amplitude Scaling and Decibels (dB)

Since amplitude scaling is such a common real-world operation (volume knobs, amplifier gain, attenuator loss), it is almost always expressed logarithmically in **decibels** rather than as a raw linear factor A. This is because human perception of loudness is roughly logarithmic, and because gains/losses of many cascaded stages multiply in linear terms but simply *add* in dB terms — much easier to work with.

6.1 Voltage/Amplitude Gain in dB

$$\mathrm{a(dB)} = 20 \,_{10}(A)$$

The factor of **20** (not 10) comes directly from the A^2 relationship in Section 5: since power A^2, and power-ratio dB uses $10\,_{10}(\mathrm{a})$, substituting $\mathrm{a} = A^2$ gives:

$$10\,_{10}(A^2) = 20\,_{10}(A)$$

6.2 Power Gain in dB

$$\mathrm{a}(\mathrm{dB}) = 10\,_{10}\left(yx\right) = 10\,_{10}(A^2)$$

Both formulas are consistent with each other — they're two ways of expressing exactly the same physical scaling.

6.3 Common Reference Values (Memorize These)

A (linear amplitude ratio)	Gain in dB
1 (no change)	0 dB
2 (double amplitude)	6.02 dB
0.5 (half amplitude)	-6.02 dB
10	20 dB
0.1	-20 dB
$2\ 1.414$	dB
$1/2\ 0.0$	$-$ dB

The "**3 dB point**" you'll hear about constantly in filter design refers to the frequency at which the amplitude has dropped to $1/2$ ($0.$) of its maximum value — equivalently, the power has dropped to exactly half.

6.4 Worked Example

An amplifier scales a signal's amplitude by $A = 5$. Find the gain in dB.

$$\mathrm{a}(\mathrm{dB}) = 20\,_{10}(5) = 20\ \ 0.6 = 1.\ \mathrm{dB}\ \ 14\ \mathrm{dB}$$

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7. Linearity and Superposition — Why Amplitude Scaling Is "Nice" Mathematically

Amplitude scaling is one of the two conditions required to prove a system is **linear**:

1. **Additivity:** $[x_1(t) + x_2(t)] = [x_1(t)] + [x_2(t)]$
2. **Homogeneity (Scaling):** $[A \cdot x(t)] = A \cdot [x(t)]$

If a system satisfies homogeneity, scaling the input by A scales the output by exactly the same A — no distortion of shape, no cross terms, no surprises. This is precisely why amplitude scaling is treated as such a fundamental "building block": any system built purely out of amplitude scaling and addition (and no nonlinear elements) is automatically a **Linear system**, and combined with time-invariance, an **LTI (Linear Time-Invariant) system** — the entire foundation of classical signal processing and control theory rests on this.

Important subtlety: real physical amplifiers are only *approximately* linear amplitude-scalers. Push the input signal too large, and the amplifier **saturates/clips** — beyond some threshold, the output no longer scales proportionally, and this is the point at which the amplitude-scaling model breaks down. See Section 9.

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8. Effect on the Frequency Domain (Fourier Transform Property)

Amplitude scaling has an extremely clean, direct effect in the frequency domain: it's a **linearity property**, arguably the simplest of all Fourier Transform properties.

If $x(t)$ () (Fourier Transform pair), then:

$$A \cdot x(t) \quad A \cdot ()$$

In words: **scaling a signal's amplitude in the time domain scales its entire spectrum by the exact same factor in the frequency domain.** The *shape* of the spectrum, its bandwidth, and the frequencies present are completely unaffected — only the height of the spectrum (magnitude at every frequency) is uniformly multiplied by A.

This directly follows from the linearity of the integral defining the Fourier Transform:

$$() = {}_{-}\, x(t)\, {}^{-2t}\, t$$

$$A \cdot x(t) = {}_{-}\, A \cdot x(t) \cdot {}^{-2t}\, t = A \; {}_{-}\, x(t) \cdot {}^{-2t}\, t = A \cdot ()$$

This is why, for example, turning up the volume on a song does not change its pitch, its rhythm, or the relative balance between bass/treble frequencies — it uniformly raises the entire spectrum's magnitude.

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9. Practical/Real-World Considerations

9.1 Clipping and Saturation

Real amplifiers, ADCs, and speakers have a maximum output limit. If $A \cdot x(t)$ exceeds this physical limit, the signal is **clipped** — the top and bottom of the waveform get flattened rather than continuing to scale linearly. This introduces **harmonic distortion** (new frequency components not present in the original signal), which directly violates the ideal amplitude-scaling model from Section 8 (a nonlinear operation like clipping does *not* simply scale the spectrum — it adds new frequencies).

9.2 Quantization Effects in Digital Systems

In digital signal processing, a signal is represented with a finite number of bits. Scaling a signal down ($A \quad 1$) reduces its representation in terms of quantization levels used, which can worsen the **signal-to-quantization-noise ratio (SQNR)**. Conversely, scaling up too far risks **overflow** (wrapping around or clipping in fixed-point arithmetic). This is why digital gain stages are carefully designed with **headroom** in mind.

9.3 Normalization

A very common application of amplitude scaling is **normalization** — scaling a signal so that its peak value becomes exactly 1 (or its energy becomes exactly 1). This is done by choosing:

$$A = 1 \max |x(t)| (\mathrm{amaa})$$

or

$$A = 1 E_x (\mathrm{maaa}\ E_y = 1)$$

Normalization is extremely common as a preprocessing step before feeding signals into machine learning models, audio mastering pipelines, or comparison algorithms — it removes the effect of arbitrary loudness/scale differences so that only the *shape* of signals is compared.

9.4 Automatic Gain Control (AGC)

AGC systems dynamically adjust the scaling factor A over time (in response to the signal's changing amplitude) so that the output stays within a target range — this is how microphones, radio receivers, and hearing aids maintain a consistent output level despite varying input loudness. Note: because A changes with time in AGC, technically this becomes a **time-varying, nonlinear** operation over the long run, even though at any single instant it's a simple multiplication.

9.5 Windowing (A Special Case Worth Knowing)

Multiplying a signal by a window function (t) that varies with time, $y(t) = (t) \cdot x(t)$, is a generalization of amplitude scaling where A is no longer constant but a function of time. This is heavily used in spectral analysis (e.g., Hamming, Hanning windows) to reduce spectral leakage before applying an FFT. It's worth knowing this connects amplitude scaling (constant multiplier) to windowing (time-varying multiplier) as a conceptual next step.

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10. Amplitude Scaling in Discrete-Time and Digital Systems

Everything above applies identically to discrete-time signals $x[n]$, with sums replacing integrals:

Energy:

$$E_x = \quad_{n=-} |x[n]|^2 E_y = A^2 \cdot E_x$$

Power (for periodic/power signals):

$$_x = 1 \;_{-1}\;_{n=0} |x[n]|^2 {}_y = A^2 \cdot {}_x$$

In digital audio, amplitude scaling is literally implemented as multiplying each sample (an integer or floating-point number) by a gain constant A, often itself derived from a dB value using:

$$A = 10^{\mathrm{dB}/20}$$

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11. Summary Cheat-Sheet

Concept	Formula	Key Takeaway
Amplitude scaling	$y(t) = A \cdot x(t)$	Multiplies every sample by constant A; time axis untouched
Amplification	$A > 1$	Signal magnitude increases
Attenuation	$0 \;\; A \;\; 1$	Signal magnitude decreases
Inversion	$A = -1$	Signal flips vertically, magnitude unchanged
Energy scaling	$E_y = A^2 \cdot E_x$	Energy scales with the SQUARE of A
Power scaling	${}_y = A^2 \cdot {}_x$	Same square relationship
Voltage/amplitude gain in dB	$20 \;_{10}(A)$	Factor of 20 because power A^2
Power gain in dB	$10 \;_{10}(A^2)$	Equivalent to the above
3 dB point	$A = 1/2 \;\; 0.0$	Power drops to half

Concept	Formula	Key Takeaway
Fourier Transform property	$A \cdot x(t) \quad A \cdot ()$	Spectrum shape unchanged, only scaled in magnitude
Linearity requirement	$[A \cdot x(t)] = A \cdot [x(t)]$	Homogeneity condition for LTI systems
Peak normalization	$A = 1/\max x(t)$	Rescales peak to exactly 1

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12. Practice Questions (Increasing Difficulty)

Q1 (Very Basic). A signal $x(t)$ has a peak amplitude of 6. If it is scaled by a factor of $A = 0.5$, what is the new peak amplitude? Is this amplification or attenuation?

Q2 (Basic). A signal $x(t)$ has energy $E_x = 25$ units. Find the energy of the scaled signal $y(t) = 4 \cdot x(t)$.

Q3 (Medium). An amplifier has a voltage gain factor $A =$. (a) Compute the gain in dB. (b) If the input power is 2 , compute the output power, and separately verify your dB answer using the power-based dB formula $10\ _{10}(_y/_x)$.

Q4 (Hard). A signal $x(t) = \ (2 \cdot 100t)$ is passed through a system that outputs $y(t) = -2 \cdot x(t)$. (a) Write $y(t)$ in the standard cosine form with an explicit phase term, showing how the negative sign corresponds to a 10 phase shift. (b) Compute $_x$ and $_y$ for these sinusoids (recall for a cosine of amplitude , average power is $^2/2$), and confirm the $_y = A^2 \cdot _x$ relationship numerically.

Q5 (Very Hard). A digital audio signal is normalized so that its peak sample value is exactly 1.0 (using peak normalization, $A = 1/\max|x[n]|$). The original signal's maximum sample magnitude was 0.25, and its original energy was $E_x = 4$. (a) Find the scaling factor A used for normalization and the new energy E_y after normalization. (b) The signal is then passed through an amplifier with a gain of $+6\ \mathrm{dB}$. Convert this dB value to a linear amplitude factor, apply it to the already-normalized signal, and check whether the resulting peak value would cause clipping if the system's maximum allowable output is 1.0. (c) Propose, with justification, the maximum gain in dB that could be applied to the normalized signal without causing clipping.